

# MODULE 7

## MATH INSTRUCTIONS AND DATA TYPE CONVERSION

PLC Siemens TIA Portal Course

Module 7: MATH INSTRUCTIONS AND DATA TYPE CONVERSION

Author: AUTOMATION AND ROBOTICS ENGINEER

Level: Basic-Intermediate

Software: Siemens TIA Portal

Language: Ladder Diagram (LAD)

## Introduction

In modern industrial automation, programmable logic controllers (PLCs) not only execute logic sequences but also perform mathematical operations to control processes, calculate production parameters, generate setpoints, and process information coming from sensors and industrial instruments.

Math instructions make it possible to perform arithmetic operations within the program, while data type conversions make it possible to work with different numeric formats safely and efficiently.

In this module you will learn to use the addition, subtraction, multiplication, and division math instructions, as well as the data conversion functions available in Siemens TIA Portal V20.

## Importance of Math Operations in PLCs

Math operations are used in practically every industrial application:

- Scaling analog signals.
- Calculating speeds.
- Unit conversion.
- Production totalizers.
- Material dosing.
- Flow calculation.
- Process control.
- Energy efficiency.
- Calculating operating times.

## Numeric Data Types in Siemens

Before using math instructions, you need to know the most commonly used data types in TIA Portal.

| Data type | Size    | Approximate range |
|-----------|---------|-------------------|
| Byte      | 8 bits  | 0 to 255          |
| Int       | 16 bits | -32,768 to 32,767 |
| DInt      | 32 bits | ±2,147,483,647    |
| Real      | 32 bits | Decimal numbers   |
| LReal     | 64 bits | High precision    |

## **ADD Instruction (Addition)**

### **Definition**

The ADD instruction adds two or more numeric values and stores the result in an output variable.

### **Operation**

`Result = Operand1 + Operand2`

### **Industrial applications**

- Production totalizers.
- Parts accumulation.
- Summing energy consumption.
- Cycle counting.
- Inventory calculation.

### **Practical example**

A production line manufactures:

`Shift 1 = 350 parts`

`Shift 2 = 420 parts`

Result:

`Total_Production = 770 parts`

## **SUB Instruction (Subtraction)**

### **Definition**

The SUB instruction obtains the difference between two numeric values.

## Operation

`Result = Operand1 - Operand2`

## Industrial applications

- Inventories.
- Remaining parts.
- Amount of material available.
- Deducting defective production.
- Storage systems.

## Practical example

Initial stock:

500 parts

Parts used:

120 parts

Result:

Remaining stock = 380 parts

## MUL Instruction (Multiplication)

### Definition

The MUL instruction multiplies two numeric variables.

### Operation

`Result = Operand1 × Operand2`

## Industrial applications

- Production calculation.
- Unit conversion.
- Signal scaling.
- Flow calculation.
- Material dosing.

## Practical example

A machine produces:

20 boxes/hour

Over:

8 hours

Result:

Daily production = 160 boxes

## DIV Instruction (Division)

### Definition

The DIV instruction divides one numeric value by another.

### Operation

`Result = Operand1 ÷ Operand2`

## Industrial applications

- Averages.
- Yields.
- Speed calculation.
- Production efficiency.
- Unit conversion.

## Practical example

Total production:

480 parts

Production time:

8 hours

Result:

60 parts/hour

## Important Considerations for Division

Before performing a division you must verify that:

`Divisor  $\neq$  0`

Dividing by zero will generate a runtime error.

# Data Type Conversion

## Definition

Data type conversion allows a variable to be transformed from one numeric format to another so calculations and operations can be performed correctly.

## Why is it necessary?

In industry it is common to find:

- Integer analog inputs.
- Decimal variables.
- Integer counters.
- Values coming from instruments.

These values often need to be converted before they can be used.

## Most Commonly Used Conversion Instructions

| Conversion   | Description           |
|--------------|-----------------------|
| INT_TO_REAL  | Converts Int to Real  |
| REAL_TO_INT  | Converts Real to Int  |
| DINT_TO_REAL | Converts DInt to Real |
| REAL_TO_DINT | Converts Real to DInt |
| WORD_TO_INT  | Converts Word to Int  |
| INT_TO_DINT  | Converts Int to DInt  |

## Practical Conversion Example

A temperature transmitter outputs:

250

Data type:

INT

To perform calculations with decimals, it is converted to:

REAL = 250.0

using:

INT\_TO\_REAL

## Scaling Analog Signals

One of the most important applications of math instructions is scaling.

Example:

Analog input:

0 - 27648

Sensor range:

0 - 100 °C

The math operation converts the read value into an actual process temperature.



# Industrial Applications of Math Instructions

## Product dosing

Calculating material quantities.

## Speed control

Converting RPM and linear speed.

## Industrial production

Calculating parts per hour.

## Weighing systems

Converting analog signals to kilograms.

## Energy efficiency

Calculating consumption and performance.

## Programming Best Practices

- Use the appropriate data type for each variable.
- Avoid memory overflows.
- Check for division by zero.
- Document the conversions performed.
- Use REAL variables for calculations with decimals.
- Verify the variable range before performing conversions.
- Test calculations by simulating in PLCSIM.

## **Practical Exercise**

Design a program that allows you to:

### **Exercise 1**

Add the production of two lines and store the result.

### **Exercise 2**

Calculate the remaining parts of a batch.

### **Exercise 3**

Calculate the hourly production of a machine.

### **Exercise 4**

Scale an analog input from 0–27648 to a range of 0–100 °C.

### **Exercise 5**

Convert an INT variable to REAL to perform calculations with decimals.

## Conclusion

Math instructions and data type conversions are fundamental tools in Siemens PLC programming. Applying them correctly makes it possible to develop more accurate, robust, and professional programs, facilitating signal processing, industrial calculations, and efficient information processing.

Mastering these instructions is essential for any Siemens PLC programmer who wants to implement advanced industrial automation solutions using TIA Portal V20.

---

**END OF MODULE 7**

**AUTOMATION STUDIO X**

PLC Siemens TIA Portal Training System